

Java Programming Internship

Project 1 Report

Project Title

Number Guessing Game

Submitted By

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Organization

Decode Labs

Objective

The objective of this project is to create a Java-based console application that generates a random number between 1 and 100. The user is required to guess the generated number. The application provides feedback indicating whether the guessed number is higher or lower than the target number until the correct answer is entered.

Problem Statement

Develop a Number Guessing Game using Java where:

- * The computer generates a random number.
- * The user enters guesses.
- * The program provides hints after every guess.
- * The game continues until the correct number is guessed.

Software Used

- * IntelliJ IDEA
- * Java JDK
- * Windows 11

Technologies Used

- * Java
- * Scanner Class
- * Random Class

Concepts Used

- * Variables
- * Conditional Statements
- * While Loop
- * User Input

* Random Number Generation

* Input Validation

Algorithm

1. Start the program.
2. Generate a random number between 1 and 100.
3. Ask the user to enter a guess.
4. Compare the guessed number with the generated number.
5. If the guess is higher, display Too High.
6. If the guess is lower, display Too Low.
7. Repeat until the correct number is guessed.
8. Display the total number of attempts.
9. End the program.

Flowchart

Start

|

Generate Random Number

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User Enters Guess

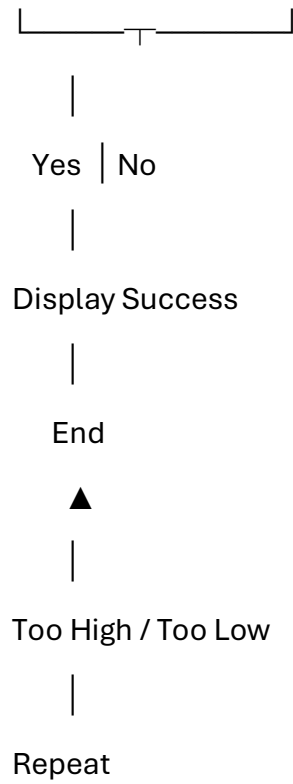
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Compare Guess

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| Guess Correct? |



Expected Output

```
=====
NUMBER GUESSING GAME
=====
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Guess a number between 1 and 100.

Enter your guess: 50

Too Low!

Enter your guess: 75

Too High!

Enter your guess: 63

Congratulations!

You guessed the correct number.

Attempts: 3

Learning Outcomes

- * Learned how to use Java for console application development.
- * Implemented loops and conditional statements.
- * Used the Random class to generate unpredictable numbers.
- * Accepted user input using the Scanner class.
- * Improved logical thinking and debugging skills.

Conclusion

The Number Guessing Game successfully demonstrates the use of Java fundamentals such as loops, conditionals, random number generation, and user interaction. This project enhanced my understanding of console-based application development and strengthened my problem-solving abilities. It serves as a solid foundation for more advanced Java projects during the internship.